Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

Xinya Du, Assistant Professor of Computer Science, The University of Texas at Dallas

Email: xinya.du@utdallas.edu **Website:** https://xinyadu.github.io

Program Officer: Dr. Behzad Kamgar-Parsi, Code 311 — Artificial Intelligence / Machine Learning, Reasoning and Intelligence

Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Intelligent and Collaborative Agents

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Problem. Autonomous agents retrieve knowledge, interpret images, use tools, and reason through multistep plans. An unnoticed error can invalidate an otherwise plausible trajectory: a resupply agent may read the quantity from the wrong row of a photographed inventory sheet and pass it to a planning tool. Outcome rewards reveal only that the plan failed; generic self-critique may miss the original visual error. Reliable agents need to identify the faulty reasoning step and choose an appropriate correction.

Central hypothesis. We hypothesize that typed, evidence-linked self-assessment improves both inference-time recovery and learned reasoning policies. Agents that localize why a step fails should outperform outcome-only learning and scalar process rewards on unseen tasks at matched cost. By distinguishing errors in observation, inference, and action, agents can select targeted corrections and recognize when further attempts are unlikely to help.

Research plan.

- **Develop neuro-symbolic representations of evidence-grounded agent reasoning.** Connect neural interpretation of text, images, and tool observations to structured representations of entities, events, preconditions, and inferential dependencies. Explicitly track alternative interpretations and provenance so symbolic checks can distinguish unsupported observations from invalid deductions or actions. Investigate when evidence and inference constraints can expose inconsistencies despite uncertain perception, and how to maintain valid conclusions as observations or external states change.
- **Learn fine-grained self-assessment of agent reasoning.** Develop methods for agents to evaluate their own multistep reasoning and actions, identifying where errors occur and why. Unlike conventional process reward models that assign a correctness score to each step, our approach will provide interpretable feedback that distinguishes errors in knowledge retrieval, visual understanding, logical reasoning, and tool use. We will investigate how to train such self-assessment models with limited error annotations and how well they generalize to longer trajectories, unseen tasks, and unfamiliar environments.
- **Turn diagnosis into agent self-improvement.** Condition the next action on the diagnosed failure: re-examine an image region, retrieve a missing fact, repair a tool argument, backtrack, or abstain. Use error-aware process feedback and successful corrective trajectories to train improved policies, while checking final outcomes independently to detect reward exploitation. Compare type-conditioned correction training against scalar process rewards, and test whether reducing one error type causes new failures elsewhere. Measure improvement across training rounds as well as immediate repair success, including cases where repeated self-correction wastes resources.

Evaluation and expected outcomes. Evaluate on MEQA, GAIA, and τ-bench, supplemented by controlled trajectories with independently verified error types and outcomes. Measure diagnosis accuracy and calibration, corrective-action success, and final task performance relative to tool cost. Compare with outcome-only RL, scalar process rewards, and LLM self-critique at matched computational budgets, separating diagnostic quality from downstream recovery.

Naval relevance. A Naval logistics support agent may combine a photographed inventory list, written guidance, and a resupply-planning tool. If it reads the wrong row, a useful self-assessment identifies the suspect image region and prompts a targeted correction before the tool result is trusted. This addresses the program's interests in uncertain reasoning and self-assessment, visual understanding, and the integration of knowledge, learning, and planning.

Prior work and PI qualifications. Our FG-PRM (Li et al., 2025, Findings of EMNLP) and HiPRAG (Wu et al., 2026, ICLR) develop process-level supervision for reasoning and retrieval. FaithScore (Jing et al., 2024, Findings of EMNLP) and FGAIF (Jing and Du, 2025, TMLR) provide starting points for multimodal feedback. The PI holds a 2024 NSF CAREER award, received a 2023 Amazon Research Award for process-guided fine-tuning, and, as a postdoc, led system integration for the RESIN team's RESIN-11 delivery system in DARPA KAIROS.